Virtual University of Pakistan

STA-100

General Mathematics & Biostatistics

| Past PApers (Till mids)|

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1) *Cos(-0)=?*

a) Cos0 *b) –cos0* *c) Sin0* *d) –Sin0*

2) The smallest interval for which the fuction repeats itself is called:

a) Peroidic function b) Period c) Cosin function d) Sine function

3) The range of cosin function is between:

a) -1 and 2 b) 1 and 2 c) -1 and 1 d) All of these

4) Sin0 is positive in _____quadrent and negative in _____quadrent:

a) 1st , 2nd B) 3rd , 4th C) 2nd , 3rd d) 1st and 2nd, 3rd and 4th

5) Sin0=

a) PQ/OP b) OP/PQ c) P/Q d)O/P

6) Tan0 =

a) PQ/OQ *b) x/y* *c) y/x* *d)* Both a and c

7) Tan0 is positive in _____quadrent:

a) 1st and 2nd b)1st and 3rd c)1st and 4th d)2nd and 3rd

8) Tan(180 -)Tan0 *b) –tan0* *c)* Both d)None of these

9) The hight from the mean value of the function to its maximum or minimum value of a function is its:

a) Period b)Periodic function c)Range d)Amlitude

10) The range of *y=arcsin x* is :

a) [–pi and pi/4] b)[-pi and pi/2] c)[pi and pi/4] d)[Pi and pi/2]

11) The domain of *y= arctan x* is :

a) (- infinity and infinity) b)(Infinity and zero) c) (–pi/2 and pi/2) d) Both a and b

12) The inverse tangent function is defined by *y=arctan x* if and only if :

a) Tan x=x *b) Tan y=y* *c)Tan y=x* *d)Tan x=y*

13) The cosine function has period of

a) 180 *b)360* *c)240* *d) 90*

14) cosine function, has the range between:

a) -1 and 1. *B) 1and 3* *c) 1and 2* *d)-2and 1*

15) For tangent function period is:

a) 90 *b)80* *c)180* *d)260*

16) Sin(theta -180)=?

*a) Sin** *b) -sin** *c) Both* *d)Non*

An equation is not an identity is called a _____ equation:

a) Identical b)Non identical c) Conditional d)Non conditional

17) In the reciprocal identity *sec0=?*

a) 1/cos0 *b)1/sin0* *c)1/tan0* *d)1/csc0*

18) In periodic properteis tan(0+pi)=

a) Sin 0 *b)C0s 0* *c)Tan 0* *d)Sec0*

19) In periodic properteis sec(0+2pi)=?

a) Csc0 *b) Sec0* *c) Cot0* *d)Tan0*

20) *Tan 2 +1=?*

a) Sec2 *b) Sin 2* *c) Cos2* *d)*Tan2

21) *Tan(A-B)=*

a) tanA – tanB/1+tanA tanB *b) tanA + tanB/1+tanA tanB*

b) tanA + tanB/1-tanA tanB *d) tanA – tanB/1-tanA tanB*

22) *Varify that* the following Equation is an identity:

Tan t-cot t/sin t cos t=sec2 t –csc2 t

Solution:

LHS= RHS

=tan *t*-cot *t*/sin *t* cos *t*=tan *t*/sin *t* cos *t*-cot *t*/sin *t* cos *t*

=tan t. 1/sin t cos t-cot t.1/sin t cos t

=sin t/cos t.1/sin t cos t-cos t /sin t.1/sin t cos t

=1/cos2 t-1/sin2 t

=sec2 t-csc 2 t

23) *What is double angle formula?prove by sin, cos and tan0s?*

=2tanA/1-tan2A

Prove ***=sin2A=****sin(A+A)=sinA cosA+sinA cosA=2sinA cosA*

*=****cos2A=****cos(A+A)=cosA cosA-sinA sinA=cos2A –sin2A*

*=****tan2A=****tanA +tanA/1-tanA tanA=****2tan A/1-tan2A***

=2tan A/1-tan2A proved

24) *To solve tan0 = k*

Ans:

___find tan-*1 (k)=0*

___use the peroidic property tan (*180+-0)=tan(0*) to get the other solution in the required interval.

25) Given that *tan(63.43 degree)=2, solve tan0 = -2*, giving all the roots in *0<0<360?*

Ans : Find $\tan^{-1}(-2) = -63.43^\circ$, Note that -63.43° is not in our range. Now use the periodic property $\tan(-63.43^\circ) = \tan(-63.43^\circ - 180^\circ)$ to get 116.56°. Now use the periodic property $\tan(116.56^\circ) = \tan(116.56^\circ + 180^\circ)$ to get 296.56°. The solutions are is 116.6° and 296.6°

26)what is double angale formula?

- Double Angle Formulas:

- sin2A=sin(A+A)=sinAcosA+sinAcosA = 2sinAcosA
- 22cos2A=cos(A+A)=cosAcosA-sinAsinA = cosA-sinA
- 2tanA+tanA2tanAtan2A==1-tanAtanA1-tanA

27)What is sum of difference formula?

- Sum and Difference Formulas:

cos(A-B) = cosAcosB + sinAsinB cos(A+B) = cosAcosB - sinAsinB

sin(A+B) = sinAcosB + cosAsinB sin(A-B) = sinAcosB – cosAsinB

- tantantan()1tantan*ABABAB*□□□□

tantantan()1tantan*ABABAB*

***28) what are* Periodic Properties ?**

ans:

$$\sin(\theta + 2\pi) = \sin\theta \qquad \csc(\theta + 2\pi) = \csc\theta$$

$$\cos(\theta + 2\pi) = \cos\theta \qquad \sec(\theta + 2\pi) = \sec\theta$$

$$\tan(\theta + \pi) = \tan\theta \qquad \cot(\theta + \pi) = \cot\theta$$

***29) What are* Pythagorean Identities ?**

Ans: sin2(theta)+cos2(theta)=1

$$\tan^2\theta + 1 = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

30) define conditional equation?

An equation that is not an identity is called a **conditional equation.**

31) Define an amplitute?

Definition: The **Amplitude** of a function is the height from the mean (or the rest) value of the function to its maximum or minimum value.

32) Difference between periodic function & period?

Definition: The functions with the property that they keep repeating themselves are called **periodic functions**. The smallest interval for which the function repeats itself is called its **period**.

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Objective

1) **A(4,3) B(10,7); *find the mid point***
2) $4\begin{bmatrix}2 & 0\\ 4 & -1\end{bmatrix}\begin{bmatrix}8 & 0\\ 16 & -4\end{bmatrix}$
3) **If b^2-4ac<zero then *there will be* *No Solution***

Subjective

1) **You have 5 chess cards named A,B,C,D,E . In how many ways you can arrange them to make a word from them (in this question we have to find possible number of permutations) (2 marks)**

$5! = 5\times4\times3\times2\times1 = 120$

2) **Find the exact value of Cos150 without using calculator. (3 marks)**

$Cos(180^0 - Cos30^0)$

By applying angle rule

$[Cos2(90))^0 - Cos30^0]$

$-Cos(30^0)$

$-Cos30^0 = -\frac{\sqrt{3}}{2}$

3) **If f(x)=4x+1 and g(x)=x-6 then find f(x)-g(x).(2 marks)**

$f.g(x) = f(x) \times g(x)$
$f.g(x) = (4x+1)(x-6)$
$f.g(x) = [4x(x)+4x(-6)+1(2x)+1(--6)]$
$f.g(x) = [4x^2-24x+2x-6]$
$f.g(x) = (4x^2-22x-6)$

4) **Write the equation in complete square form , x^2 +4x+2 .(3 marks)**

$x^2 + bx + \left(\frac{b}{2}\right)^2 + 2 - \left(\frac{b}{2}\right)$

$x^2 + bx + \left(\frac{4}{2}\right)^2 + 2 - \left(\frac{4}{2}\right)$

$x^2 + bx + \left(\frac{16}{4}\right) + 2 - \left(\frac{4}{2}\right)$

$x^2 + 4x + \left(\frac{16}{4}\right) + 2 - (2)$

$x^2 + 4x + 4$

$(x+2)^2$

5) .(2-4i)(3+5i)

2(3+5i)-4i)(3+5i)

$6+10i-12i-20i^2$

6-2i-20(-1)

6-2i+20

26-2i

6) .a=3000, r=$\frac{1}{2}$ find the a_{10}

.a=3000, r=$\frac{1}{2}$ find the

$a_{10} = a_1 r^{10-1}$

$a_{10} = (3000)(r^9)$

$a_{10} = (3000)(\frac{1}{2})^9$

$a_{10} = (\frac{3000}{2^9})$

7) A club has five seniors and three junior players . For and upcoming tournament they have to make a team of two juniors and two seniors. how many possible number of teams they can make (this question is from combinations) (5 marks)

$${}^{r}P_s = {}^{2}P_2 = \frac{2!}{2-2!} = \frac{2!}{0!} = \frac{2*1}{0!} = \frac{2}{1} = 2$$

8) If f(x)= 3x-1 and g(x)= 2x-1 then find the product of i.e, f.g(x)

*f.g(x)= f(x) * g(x)*
f.g(x)= (3x-1)(2x-1)
f.g(x)= [3x(2x)+3x(-1)-1(2x)-1(-1)]
f.g(x)= [$6x^2$-3x-2x+1]
f.g(x)= [$6x^2$-5x+1]

9) Fg(x)
fx()= x^3+1
g(x)= 3^2+1
Same method will be used as above;

10) **Expand the *Sin(*A-B*)***

*Sin(*A-B*)* = ***(****Sin*A+(-B***))***

*Sin(*A-B*)* = *Sin*A+*Cos*(-B*)* + *Cos*A+*Sin*(-B*)*

*Sin(*A-B*)* = *Sin*A+*Cos*(B*)* + *Cos*A+*Sin*(-B*)*

*Sin(*A-B*)* = *Sin*A+*Cos*B - *Cos*A+*Sin*B

11) **Write down the properties to reduce a metric to reduce echelon form.**

Definition:

"A matrix is in ***echelon form*** *if it has the following properties"*

- *Every non-zero row begins with a 1 (called a* ***leading*** *1)*
- *Every leading one in a lower row is further to the right of the leading one above it.*
- *If there are zero rows, they are at the end of the matrix*

Example; $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

Definition:

A matrix is in ***reduced echelon form*** *if in addition to the above three properties it also has* ***t****he following property:*

- *Every other entry in a column containing a leading one is zero*

Example; $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

12) **If Z=2-B*i* find out Z.Z*=?**

Z=2-B*i* Z*=2+B*i*

Then,

Z.Z* = (2-B*i*)(2+B*i*)

Z.Z* = [2(2)+1(B*i*)-B*i*(2)-B*i*(B*i*)]

Z.Z* = [4+B*i*-2B*i*-B*i*^{2}]

Z.Z* = [4+(1-2)B-B(-1)]

Z.Z* = [4+(-1)B+B]

Z.Z* = [4-B+B]

Z.Z* = 4

13) **What is the cardianality of Sets?**

A= set of English alphabets

B= set of integers

C= set of even number between 3 and 7

A= {a,b,c,d,……,z}

Caridanility of set A is |A|= **26**

B= {0,±1, ±2, ±3,…..}

(Set B *is a infinite there for its caridanility can't be counted)*

Caridanility of set B is |B|= **is uncountable**

C= {4,6}

Caridanility of set C is |C|= **2**

14) **Find the slope of** ***2x-3y= 12***

2x-3y= 12
2x-3y-12 = 0

Slope = $-\frac{a}{b} = -\frac{2}{-3} = \frac{2}{3}$

Intercept= $-\frac{C}{b} = -\frac{-12}{-3} = -\frac{12}{3}$

15) **Find Domain & Range** ***for f(n)= 5n+2***

Domain = $\{n \in R \mid n \geq 0\}$ R, Range= R

16) **Radius of Circle** $(x-5)^2 - (y-7)^2 = 4a$

$(x-5)^2 - (y-7)^2 = 4a$

Center equation of a circle

$(x-k)^2 - (y-h)^2 = r^2$

$(x-5)^2 - (y-7)^2 = 7^2$

17) **Find the slope of** ***(2, -1)*** **and** ***(-5, 3)***

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$m = \frac{3-(-1)}{-5-2}$

$m = \frac{3+1}{-7}$

$m = \frac{4}{-7}$

18) $x^3+6x-3=0$ **,find value of using "*Quadratic equation*"**

$x^3+6x-3=0$

$x^3+6x=+3$

divide by "x" on both sides

$x^2 - \frac{3}{x}+6=0$

$$.x^2+2\frac{3}{2t}+\left(\frac{3}{2t}\right)^2=\frac{7}{t}+\left(\frac{3}{2t}\right)^2$$

$$\left(x+\frac{3}{2t}\right)^2=\frac{7}{t}+\left(\frac{9}{4t^2}\right)$$

$$\left(x+\frac{3}{2t}\right)^2=\left(\frac{28t+9}{4t^2}\right)$$

$$\sqrt{\left(x+\frac{3}{2t}\right)^2}=\sqrt{\left(\frac{28t+9}{4t^2}\right)}$$

$$.x+\frac{3}{2t}=\pm\sqrt{\left(\frac{28t+9}{4t^2}\right)}$$

$$.x+\frac{3}{2t}=\frac{-3\pm\sqrt{28t+9}}{2t}$$

19) $.tx^2+3x-7=0$,find value of "x"

$$.tx^2+3x-7=0$$

$$.tx^2+3x=+7$$

divide by "t" on both sides

$$.x^2+\frac{3}{t}=\frac{7}{t}$$

$$.x^2+2x\frac{3}{2t}+\left(\frac{3}{2t}\right)^2=\frac{7}{t}+\left(\frac{3}{2t}\right)^2$$

$$\left(x+\frac{3}{2t}\right)^2=\frac{7}{t}+\left(\frac{9}{4t^2}\right)$$

$$\left(x+\frac{3}{2t}\right)^2=\left(\frac{28t+9}{4t^2}\right)$$

$$\sqrt{\left(x+\frac{3}{2t}\right)^2}=\sqrt{\left(\frac{28t+9}{4t^2}\right)}$$

$$.x+\frac{3}{2t}=\pm\sqrt{\left(\frac{28t+9}{4t^2}\right)}$$

$$.x=\frac{\pm\sqrt{28t+9}}{2t}-\frac{3}{2t}$$

$$.x+\frac{3}{2t}=\frac{-3\pm\sqrt{28t+9}}{2t}$$

20) **A= {2,4,6,8,10,12,14,16}**
B={12,14,16,20}
find **A∩B**

A∩B= {2,4,6,8,10,12,14,16} ∩ {12,14,16,20}
A∩B= {12,14,16}

21). **a_0=2, a_1=3, a_2= -2, a_3=1**

$\sum_{2=0}^{2} 2a = 2(\mathbf{a_0}+ \mathbf{a_1}+ \mathbf{a_2})$
=2(2+3+(-2))
=2(3)
=6

22) **A=$\begin{bmatrix}1 & 2 & 1\\ 2 & 1 & 2\\ 3 & 1 & 3\end{bmatrix}$ B=$\begin{bmatrix}2 & 2 & 3\\ 2 & 1 & 1\end{bmatrix}$, is multiplication possible**

Order of A= 3 × 3
Order of B=2 × 3
For the multiplication
Number of Columns in A = Number of rows in B
Here,
3≠2
SO, multiplication is not possible.

23) **A local family Restaurant have special breakfast and a customer choose one of them each day**

Egg Chicken	Mango, Orange	Orange
Egg mutton	Apple fruit Slice	Rasberry
Egg beef	Cup fruit juice	Tomato

Find the possible combination without meet.

Since we have to make combination without meet,

$^{3}P_3 = \frac{3!}{3-3!} = \frac{3!}{0!} = \frac{3*2*1}{0!} = \frac{6}{1}$

24)x^2+6x+5= 0

$x^2+6x+5=0$

$x^2+1x+5x+5=0$

$x(x+1)+5(x+1)=0$

$(x+5)+(x+1)=0$

$(1+x)+(1+5)=0$

$(x+1)=0$ $(x+5)=0$
$x=-1$ $x=-5$

S.S={-1, -5}

25) **If 3rd and 4th terms of geometric series are 12 & B respectively. Find out the sum to infinite.**

3rd terms of geometric series

$A_n = a + (n-1)d$

$A_n = 12 + (3-1)12$

$A_n = 12 + (2)12$

$A_n = 12+24$

$A_n = 36$

4rd terms of geometric series

$A_n = a + (n-1)d$

$A_n = 12 + (4-1)12$

$A_n = 12 + (3)12$

$A_n = 12+36$

$A_n = 48$

Thank You !